

REMARKS BY DR. ARNISON.—In the first case (T. R., aged 14), the increase in severity of the symptoms was gradual; food was taken and digested; and it was hoped, from the comparatively chronic character of the attack, that recovery might take place without operation further than the amputation of the finger, until the boy's life was placed in such imminent peril from suffocation. The ulnar nerve was stretched, as well as the other two, as it was uncertain how far the injury to the ring finger might be the cause of the attack, though confined to destruction of the nail, and at the time of operation nearly well. The after-progress of the case makes it uncertain to what must be given the credit of the result. There was not much immediate benefit from the nerve-stretching, and the boy would most likely have died if Calabar bean had not been used; and when that had to be omitted, it was necessary to continue the administration of chloral for some time; but, on the other hand, until the nerves were stretched, treatment failed to have any control over the disease, and after the operation the progress to recovery began. I am inclined to think that the patient was kept alive by Calabar bean until the nerve-stretching had time to exert its influence, and that he would have died if either had been omitted.

In the second case (J. H., aged 41), the period of incubation is unusually long—forty-one days from the accident, thirty-two from the amputation. The symptoms were never so severe as in the first case, but sufficiently well marked to place the diagnosis beyond doubt. Nerve-stretching would perhaps again have been employed if there had been a clear indication which nerves to stretch; the injury to the arm was the more likely cause of the tetanus, and there was marked tenderness over the brachial plexus; but, on the other hand, tetanus may arise from a simple dislocation. The first indications of its approach were felt in the thigh, and the obturator pain was at least as prominent as the brachial tenderness. Chloral hydrate was the chief remedy employed, and to it the recovery is due. Free purgation was used in both cases.

In cutting down upon the nerves for the purpose of stretching them, the median and ulnar nerves were both found by an incision along the inner border of the biceps, the muscular spiral by an incision internal to the supinator longus.

REVIEWS AND NOTICES.

THE CHANGE OF LIFE. By E. J. TILT, M.D. Fourth Edition. London: J. and A. Churchill.

THIS well-known work still holds a certain section of the medical public, as is shown by the appearance of a fourth edition. It represents the anti-surgical school of gynecology; and, perhaps, just as some pioneers of that school err in that they interfere needlessly, so does the present work fall short in that it ignores operative measures, which have been proved, over and over again, to be of the greatest value in the cure of disease. The statement in the author's preface, referring to Dr. Emmet, as follows—"Subacute, or chronic inflammation of the cervix, which so often shows itself by uterine ulceration, is for him a myth, a delusion, and nothing but the everted rim of a more or less lacerated cervix"—gives the clue to the whole theory upon which Dr. TILT bases his work. He quotes the superficial sneer, that trachelorraphy is "une débauche chirurgicale;" so it may be, if needlessly undertaken; but then all surgical proceedings might be similarly stigmatised if performed without due reason. The work contains much valuable information on the diseases, nervous and others, which accompany the cessation of the menstrual function.

DISEASES OF WOMEN. By ARTHUR W. EDIS, M.D., F.R.C.P. Second Edition. London: Smith, Elder, and Co. 1882.

IT is not by any means an easy task to write a text-book upon any subject, in the domain of medicine, which will meet with any marked success among students and practitioners of medicine. Such a book must combine several qualities in a large degree. It must be handy, well illustrated, full of the latest advances in the theme it treats of, judiciously dogmatic, trustworthy, and, lastly, it must be readable. Dr. EDIS, it is no exaggeration to say, has succeeded, in an unusual degree, in producing a book which may honestly lay claim to these qualities. The first edition was so quickly exhausted, that but little time has elapsed between the appearance of the first and second editions, for the emendations of doubtful or the incorporation of new matter. The work, however, bears witness to having undergone conscientious revision and correction. Although he has thus revised it, the author has not increased the bulk of the book, having preferred to keep to his original intention of making it a practical manual for the student and junior practitioner, and not an encyclopædia. The work has taken a high place among the standard text-books on gynecology.

BRITISH MEDICAL ASSOCIATION. SUBSCRIPTIONS FOR 1882.

SUBSCRIPTIONS to the Association for 1882 became due on January 1st. Members of Branches are requested to pay the same to their respective Secretaries. Members of the Association not belonging to Branches, are requested to forward their remittances to the General Secretary, 161A, Strand, London. Post Office Orders should be made payable at the West Central District Office, High Holborn.

The British Medical Journal.

SATURDAY, SEPTEMBER 30TH, 1882.

THE VACCINATION INQUIRY AT NORWICH.

WHILE sympathising with Dr. Guy in the annoyance and vexation to which he was subjected, during the course of the inquiry held by a Commission, recently appointed to investigate ten cases of erysipelas following vaccination, at his public station in Norwich, we desire to weigh the evidence to hand, with the double object of tracing cause and effect, and of finding a means whereby such untoward results may in future be guarded against.

It is well that this inquiry was instituted tolerably soon after the occurrence of the events examined. Too often such an investigation has been delayed so long, that the best aids to memory furnish only imperfect or improbable data. The reports of this inquiry, which have from time to time appeared in the *Times*, furnish the basis of the following criticism. Besides being meagre, these paragraphs do not furnish any consecutive record of the dates, after vaccination, at which the erysipelatous or septicæmic sequelæ attacked their victims. The general inference suggested by the published reports is, that, in most of these ten instances, the phenomena of the blood-poisoning followed the development of vaccinia; in some instances, the former showed itself before the eighth day or date of maturation of the vaccine vesicle; in others, it occurred after this day, when the vaccine fever was at its height. In some instances, there is record of a distinct erysipelatous or erythematous rash commencing at the vaccinated spots, starting at more distant foci in others. In other instances, the published reports lead to the inference that phlebitis or septicæmia were the forms of illness to which the children succumbed. That these diseases are for the most part (though not necessarily) intimately associated with wounds, however slight, is well known to all surgeons. "When the extraordinary power of vaccine lymph is considered, it ought not to seem extraordinary (says the *Times* leader on the Norwich inquiry) that the door opened for it as a prophylactic should, in bodies waiting only a sign to fall into disorder, be occasionally perverted for hostile purposes." That various inflammatory affections of the skin, cutaneous eruptions of great variety, might accompany or follow vaccinia, was a fact well known to Dr. Jenner and his co-workers. In a letter to the *Times*, Dr. H. D. Dudgeon draws attention to this; and states that Dr. Pearson wrote to Dr. Jenner (February 15th, 1799) thus: "In my patients, the inflammation around the inoculated (*i.e.*, vaccinated) parts has been merely what I would call erythematous, or the same kind of affection as in scarlatina." Dr. Woodville says, in a letter dated 1799: "We cannot boast that our cow-pox inoculation has never been attended with eruptions."

As regards the Norwich Inquiry, we are glad to find Mr. Corrie Grant, the legal representative of the parents whose children were the sufferers, acquiescing in praise of Dr. Guy's skill, experience, and care as a vaccinator (*Times* report, September 5th). The evidence on this point was abundant and convincing; so Mr. Grant proceeds to examine five causes which might lead to the outbreak of disease among vaccinated children. These he enumerates thus:—Circumstances connected with (a) the vaccinifers, those connected with (b) the mode of vaccination, with (c) the vaccinated patients, with (d) atmospheric or epidemic conditions, with (lastly) the vaccine lymph. The conclusion

he arrived at, after careful consideration of the evidence adduced, was "that the presence of erysipelas in the cases under inquiry appeared to be associated with the last of these causes, the lymph." Yet we find it reported in the *Times*, September 1st, that "a child named Johnson, which was vaccinated from Mrs. Lambert's child, had also not experienced any ill effects." Mrs. L.'s child was one of the cases under investigation, and yet more than one child had received the lymph from this vacciner with impunity. Further, from the same report we learn that other children vaccinated from the child Carver, who was vaccinated from Sewell (along with Lambert) had *not* suffered. No wonder then that Dr. Guy's suspicions were not aroused when some of these children were afterwards brought to him in a feverish state of erythema. The logical acumen of the leading legal counsel on the opposite side, can find no other source for these untoward results, than the very vaccine lymph which proved harmless when introduced into the systems of other children placed under circumstances exactly similar with those who had suffered. If it were true, as stated by Mr. W. J. Collins, M.B.M.S. (Lond.) in a letter to the *Times*, dated September 7th, that "in vaccination we inoculate a septic animal fluid, whose mode of action is unknown to us, and whose effects we cannot measure, and no amount of care or caution can obviate results like those which have provoked pardonable indignation at Norwich," it is indeed a most grave responsibility which the Governments of all civilised nations have undertaken, on the earnest entreaty of the medical profession, in legally enforcing vaccination. Vaccine is *not* a septic fluid, although we know as little of its motor power as we do of that of castor oil.

"What concerns the public is much less whether a particular practitioner has been careful or careless, than the relation between the disasters which have befallen ten of his patients, and a principle as well as a system" (*Times*, September 4th). The principle stands on a firm basis of nearly a century's experience of vaccination all over the world. The system of public vaccination, as conducted in this country, may probably with advantage be subjected from time to time to close scrutiny. The directions given to public vaccinators by the Local Government Board are explicit and useful. They require him to look at the child before vaccinating it, to see that it is healthy, and presents no cutaneous eruption. Further, it is suggested, when possible, that any symptoms of ill-health, as diarrhoea, or any concomitant unfavourable conditions, as the presence of infectious disease in other members of the family, should be inquired into before vaccination is performed. In the next place, the public vaccinator is instructed to remove lymph only from vesicles which pour out freely, and which are not abraded nor are surrounded by areola. He is cautioned not to remove with the vaccine, blood, epithelial scales, purulent-looking fluid, or any contaminations of the lymph. He is, further, recommended to wash after every second vaccination the instrument with which he operates. Following closely these directions, it is scarcely possible for him to err. Parents are only too ready to dilate on symptoms presented by their children; while the mottled skin and firm feel of a child's arm, with the absence of any eruption on the scalp, will at a glance satisfy the public vaccinator as to the child's health. The clear fluid rapidly bubbling up from vesicles should alone be used. But here we meet with a difficulty. Public vaccinators are only very poorly remunerated for the time and care bestowed on their duties; but they are required to make bricks without straw. They are afforded only partial opportunities for obtaining lymph, as compared with other practitioners. On one day they may have twenty vaccinifers and only twelve to vaccinate, while, on another occasion, they may have four vaccinifers, of whom only two present the bubbling crystal springs, while there are twenty to be vaccinated. Surely the only disease which is known to avert another disease, ought to be consigned to the custody of specialists. If large vaccination areas were arranged, and entrusted to the charge of medical men who devoted exclusive attention to vaccination, not only could an unlimited supply of trustworthy humanised vaccine be thus obtained, but this vaccine could readily be rejuvenated through the calf.

Now, could we, by employing animal vaccine alone, avert the possible advent of erysipelas and cutaneous diseases? Our reply must be in the negative. We can multiply our lymph supply to almost any extent by means of animal vaccination, but we cannot overlook its drawbacks. No animal disease can be communicated to man through vaccine lymph (as far as observation has hitherto gone); but "an outbreak of erysipelas has often enough followed the employment of an absolutely pure vaccine lymph from a healthy child or calf." "The infectious diseases of wounds, and especially erysipelas, are not dependent on the source of the lymph." (P. Boëmer—"Neben Vaccination mit Animaler und Humanister Lymph unter besonderer Berücksichtigung der Impfersipels;"—in *Deut. Med. Wochen.* No. 5, 1882). On the other hand, the advocates of animal vaccination assert that erysipelas, and like ill-results, follow the employment of calf vaccine much more seldom than they do vaccination with humanised vaccine. We admit this, but it may be explained otherwise than through the purity of the lymph. Before removing lymph from the calf, great care is taken that the surface of the vesicles be perfectly clean. Now, equal care should be bestowed on the human subject. Many parents, in all classes of society, are under the impression that after an arm has been vaccinated it should not be washed till the lymph has been removed. Other parents, especially primiparæ, are readily induced to apply cold cream or lard around, or poultices over, the vesicles for a day or two before they bring their children for inspection, to allay the consequent inflammation. It is, therefore, no uncommon thing for a public vaccinator to meet with a vacciner showing a greasy, unclean arm, which he ought to purify before puncturing the vesicles. But even if such applications have not been used before the infant was brought for inspection, they are very commonly applied immediately after, and the lymph oozing over such an uncleanly surface soon becomes poisonous. It would be well, therefore, if public vaccinators advised parents after vaccination to wash off the lymph oozing from the vesicles and dry the arm, but on no account to apply cold cream, ointments, etc. Another common excitant of unhealthy inflammation in vaccination vesicles is the access of worsted, threads, from clothing, and especially of aniline dyes from coloured ribbons used to keep up the sleeves in infants. These aniline dyes are known to induce eczematous and other eruptions when they come in contact with sound adult skins, and they no doubt frequently cause irritation when partly dissolved and applied to abraded vaccine vesicles.

It would, indeed, be a happy outcome of the Norwich inquiry if thereby the Local Government Board were led to reconsider the present arrangements for public vaccination, especially in respect to the efficient carrying out of the arm-to-arm method, and the supply of lymph, as also the proper remuneration of the vaccinators.

As regards the antivaccinators, who doubtless will make hay while the sun shines, and will suck honey from the Norwich inquiry, it may not be inapt to quote from the preface by Aculeus to his letters published in 1805: "If the opposers of this benign discovery will obstinately and pertinaciously persist in their clamorous hostility, it is in vain to expect they will be diverted from their purpose by argument; no accumulation of evidence, however satisfactory and conclusive, will compel them from their ground, which is altogether independent of the real merits of cow-pox, and centres in private interest in opposition to public good."

THE HUDDERSFIELD LEAD-POISONING CASE.

AN action for damages for injury resulting from the use of a public water-supply has recently been decided at the Yorkshire Summer Assizes against the Huddersfield Corporation. The case, which is a notable one, has not attracted that public attention which it merits from its scientific bearings and its instructiveness. We are indebted to the *Chemical News* of August 25th last for an account of the principal facts of the case. The well known scientist, Mr. Crookes, the editor of that journal, having been professionally engaged on the trial, we may assume that the report in the *Chemical News* states the proved scientific facts and opinions expressed with accuracy.

A solicitor practising in Huddersfield, and living in the outskirts of that town, was the plaintiff in the action, and recovered damages. About a year ago, he was attacked, first with violent colic, and ultimately with wrist-drop, and the other usual symptoms of lead-poisoning. He lost almost entirely the use of his arms; his brain was affected; and at one time his life seemed in peril. He has subsequently, under medical treatment, partially recovered the use of his limbs; but at the trial it was stated that he was still unable to dress himself. The fact that the plaintiff was suffering from lead-poisoning being beyond all dispute, the question arose as to how the noxious metal was introduced into his system. It appears that the water-supply, which, as in most of the large northern manufacturing towns of the North of England, is the property of the corporation, is derived from several reservoirs, the plaintiff's house being served from the Blackmoorfoot reservoir. All the various water-sources are remarkably free, both from saline constituents and organic impurities. They are consequently exceedingly soft, and that from the Blackmoorfoot dam is perhaps the purest. Unfortunately, in consequence of that very purity, it is, though perfectly wholesome in reservoirs, channels, and mains, capable of acting upon leaden service-pipes. Concerning the immediate cause and *modus operandi* of this action, differences of opinion prevail; and to these differences of opinion we shall again advert.

In samples of water drawn from the plaintiff's house, Mr. Jarman, the borough analyst for Huddersfield, found 0.34 grain of lead per gallon; whilst Mr. Fairley, of Leeds, found 0.4 grain per gallon. Mr. A. Allen, of Sheffield, found a much larger quantity, viz., in two samples respectively, 0.77 and 0.84 grain of lead per gallon. On the other hand, Dr. Tidy found, in a sample of water from the plaintiff's kitchen, only 0.01, and in that from the washhouse 0.04 grain of lead per gallon. As the reporter of the case justly observes, these discrepancies are quite intelligible, without casting any reflection upon the accuracy of the analyses. The contamination with lead being derived from the service-pipes which the water traversed on its way from the mains, it is plain, he says, that different samples must vary largely, according to the time the water had been allowed to stand in them before being drawn. We presume, from this remark, that the water was not stored in leaden cisterns. It is not creditable, however, that samples for analysis should have been drawn under such widely different circumstances as to cause a variation in the amount of noxious metal present from 0.01 to 0.80 grain, without some statement being made as to the conditions under which the samples were drawn.

We now come to the most important points from a public health point of view. Does the water of the Blackmoorfoot dam dissolve lead more rapidly than soft waters generally do; and, if so, why? Again, why do soft waters so readily dissolve lead? Mr. Jarman, in an official report presented to the Huddersfield Corporation in 1879, states that the Blackmoorfoot water very closely resembles that supplied by the Manchester Corporation. Indeed, the waters of Halifax, Bradford, Leeds, Batley, Sheffield, and Manchester, as well as that from most of the Huddersfield reservoirs, if allowed to lodge in leaden pipes for twelve hours, all take up appreciable quantities of the metal. Dr. Robertson is even reported to have said that the Manchester water has been found to contain 0.3 grains of the poisonous metal per gallon. Hence it must be concluded that the Blackmoorfoot water does not form any marked exception to other mountain waters. Nevertheless, we are surprised to learn that the Manchester water-supply ever contained so much as 0.3 grain of lead per gallon; and we imagine that no waters could possibly contain so much of that dangerous metal habitually, and yet be as a rule unproductive of toxic results. We believe the rule is a good one, that any water containing one-tenth of a grain of lead per gallon, or even one part per million, should be rejected as unfit for drinking purposes.

It may be asked why, if lead be so generally present, or liable to be present, in the pipe waters supplied to our northern cities, is lead poisoning not more common? In the Huddersfield trial, the testimony of the medical witnesses was that such cases are not rare; and that

lead is exceedingly capricious in its action; and it is stated that an individual's power of resisting the influence of lead, or, indeed, of many other poisons, stands in no perceptible relation to his apparent health and vigour.

The Huddersfield water has an acid reaction, and this has been shown to be due to the presence of sulphuric acid, or an acid sulphate. As to the precise action of this acid, there was a wide difference of opinion among the experts. Drs. Crakes, Odling, and Tidy were of opinion—an opinion which we believe to be highly questionable, and resting upon no sufficient experimental data—that sulphuric acid, if present in small quantities, must tend to protect the pipes from the action of the water, by forming a thin layer of the insoluble, or, at least, very sparingly soluble, lead sulphate. Mr. A. Allen of Sheffield took a different view: one which we are surprised to find the editor of the *Chemical News* described as “novel.” Mr. Allen said that sulphuric acid would cause corrosion; and the fact that Mr. Jarman recommended as a remedy the addition of lime-water, tends to show that he, too, shared Mr. Allen's views on this point. We hope the point will be the subject of special experimental observation. It is a notorious fact that the simultaneous presence of free acid and free oxygen in water is a condition eminently favourable to the oxidation of the metal; and this condition was existent in the Blackmoorfoot water. That the lead sulphate thus formed did not prevent deleterious quantities of lead going into solution, is also incontestably proved by the large quantities of lead found by three analysts. Lead sulphate, though sparingly soluble, is, in fact, not altogether insoluble; and in this case, the lead sulphate spoken of as coating the interior of the conduit pipes—whether existent or not—was clearly incompetent to protect the water from contamination with lead. We, at the same time, freely admit that Dr. Tidy's fears that the use of an excess of lime might increase the mischief, may be well founded; and the use of lime-water to neutralise the free acid of the town supplies ought not to be permitted till its use has received ample experimental investigation.

The origin of the acidity of the Huddersfield water is curious, and is supposed to be this. The Blackmoorfoot dam receives certain ferruginous springs. These ferruginous compounds are, we presume, of the nature of ferrous salts (proto-salts). The iron salts thus introduced, on being diffused in the water and exposed to air, become split up into a basic salt, which is deposited; and free acid, or, perhaps, an acid ferric salt (persalt of iron), which remains in solution. If, as was suggested at the trial, the frequent action of soft pure waters upon boiler plates, as at Glasgow, is due to the corrosive action of oxygen held in solution in the water, the prevention of the corrosion of lead pipes by soft waters presents a great difficulty, since de-aeration of the water, even if practicable and advisable from a dietetic point of view, would render the water unpalatable. This question of the action of soft pure water upon lead, is one of extreme interest, in view of the extensive use of these waters by large towns, and their proposed further use for the metropolis.

We are greatly indebted to the reporter in the *Chemical News* for our knowledge of this interesting case; and we have borrowed largely from his report, using his own words freely. We venture, however, to express our dissent from some of his conclusions.

PARASITIC HÆMOPTYSIS AND DISTOMA RINGERI.

A RECENT number of the *Medical Reports* of the Chinese Imperial Customs, a serial that is known in Europe chiefly through the fact that it has been the medium chosen by Dr. Manson of Amoy for the publication of some of his best work, contains a contribution by that investigator on a parasite concerned in the production of hæmoptysis. The title of the paper is “Distoma Ringeri and Parasitical Hæmoptysis.” Dr. Manson refers to a previous paper published in the *Reports*, in which he had called attention to a new parasite, the mature form of which had shortly before been discovered by Dr. Ringer in Tamsui, Formosa. In his previous paper, the author had succeeded

in associating this animal with a peculiar form of recurring hæmoptysis common in one part, at least, of the Chinese Empire, and which had not previously been understood. At the same time, Professor Baelz of Tokio had been working at the subject in Japan, in which country he had found this parasitic disease to be not uncommon. The bodies described by Professor Baelz as gregarinæ, were proved by Manson to be the ova of *Distoma Ringeri*. In making a *post mortem* examination of a Portuguese who had died of aneurysm of the aorta, Dr. Ringer had found a parasite in the lungs; in the sputum of a Chinaman suffering from a chronic intermitting hæmoptysis, Dr. Manson had found certain bodies which he had no difficulty in recognising as the ova of a parasite; and when these bodies and the ova emitted by *Distoma Ringeri* were compared, they were found to be identical in size, shape, colour, and contents. In characteristic sputum sent to Dr. Manson by Professor Baelz from Japan, it was seen that the bodies described by the latter observer were identical with the ova of *Distoma Ringeri*.

The geographical distribution of this parasitic hæmoptysis seems to be limited. At Amoy it is not common, Dr. Manson having failed to find the parasite in the sputa of about 150 individuals who resided in that district. It is quite otherwise in North Formosa, which is separated from Amoy by about 200 miles of sea. Dr. Manson's application to a resident of Tamsui, a port of Formosa, was at once answered by the despatch of two bottles full of ova-laden sputum, obtained from his two servants, both of whom suffered from parasitic hæmoptysis. The disease was reported to be extremely common in that locality. Dr. Manson had, at a former period, been struck by the large number of cases of hæmoptysis he had met with in Central and South Formosa, and has now little doubt that in *Distoma Ringeri* we have the explanation of their frequent occurrence. The prevalence of the parasite in Japan and Formosa, and its apparent absence on the mainland of China, lead him to speculate on the possible existence of an intermediary host, which finds the conditions necessary for its life in soil or geological structure common to Japan and Formosa, but not present on the neighbouring continent.

Parasitic hæmoptysis can readily be diagnosed. There is a history of irregular intermitting hæmoptysis associated with slight cough, and, in the intervals of more active bleeding, the expectoration once or several times a day of small pellets of viscid brownish mucus. Violent exercise is apt to produce profuse hæmorrhage; and irritation of the lung in any way, so as to induce coughing, causes the discharge either of quantities of blood or of the characteristic sputum. At the same time, there are no objective symptoms of lung-disease, and the patient probably enjoys good general health. Examination of a small portion of the sputum with a microscope at once settles the diagnosis.

Dr. Manson gives the histories of two cases, one of which may be taken as illustrative of the course of the disease. A male, aged 31, began to spit blood eleven years previously, when working on the tea-hills near Banka. At first he noticed, when he breathed hard in carrying heavy burdens, that he coughed a little and brought up mucus mixed with blood; from that time onwards, he spat blood more or less constantly—some days none, other days a considerable quantity. Once, when pulling in a boat, about two years before he was seen by Dr. Manson, he suddenly brought up over a bowlful of pure blood; but as a rule, unless when exerting himself violently, he only brought up a few drops mixed with the mucus. Sometimes he did not spit for a few days, perhaps a month together, and then the hæmoptysis recurred, to last for one or two months. He had a slight cough; but, on auscultation, nothing much amiss could be detected. His thorax was very finely developed. He had been in the habit of drinking water from river, well, paddy-field, or ditch, whichever lay most convenient; and he said that all the inhabitants of North Formosa are similarly indiscreet.

When examined with the microscope, the ova of *Distoma Ringeri* are seen to be shaped very much after the fashion of a fowl's egg, with the exception that a circular operculum, about half the breadth of the egg, closes the broad end. On an average they measure about $\frac{1}{16}$ " $\times$ $\frac{1}{32}$ "

but some specimens are slightly larger and others slightly smaller. Their colour, which, when blood is entirely absent, as is sometimes the case, imparts the characteristic brownish tinge to the sputum, is a dirty reddish brown, and appears to reside both in the shell and in the granular portion of its contents. The shell is without markings, and shows in double outline, more especially when it has been fractured by pressure.

Dr. Manson found that the ova from the sputum developed in filtered well-water, and was thus able to trace the embryo through its various stages until it emerged from its shell, when it could be seen moving actively about by ciliary motion in the field of the microscope. He summarises his observations and views thus. The ova are laid into the bronchia mucus; in the sputum, they are cast on the ground; by rain or other means they are carried to stagnant water; they sink to the bottom; in the course of six weeks or two months, ciliated embryos are developed; when mature, these force their opercula and swim free in the water. What the next stage may be, can only be conjectured. Doubtless they enter the body of some fresh-water animal to undergo further metamorphoses. Perhaps this animal is eaten by man, or possibly the parasites once more obtain their freedom, and, while still in the water, are swallowed, and thus obtain an opportunity of gaining access to the human lungs—their final destiny.

Many points connected with this parasite still wait an explanation; and we have little doubt that workers in China and Japan will be able, before long, to add to the information which has been given by Dr. Manson. A more widespread acquaintance with his observations will, doubtless, stimulate observers in other parts of the world to investigate the possible connection of parasitism with obscure forms of disease.

DURING the past week, six deaths from scarlet fever have been registered at Accrington, where a number of new cases of the disease have been reported. At Whitchurch, Salop, scarlatina, which has been prevalent for some weeks, still continues to spread.

MRS. NATHANIEL MONTEFIORE has given £2,600 to University College Hospital, for the endowment of two beds, in memory of her late brother, Sir Francis Henry Goldsmith, Bart., and her late son, Mr. Leonard Montefiore.

IN the year ending October 31st last, there were 114 orders of inquiry in commissions of lunacy executed by the Masters in Lunacy, and the percentage on lunatics' incomes under general orders amounted to £27,060. There were 234 petitions presented for hearing.

IN the year ending September 29th, 1881, the number charged and detained as criminal lunatics in England and Wales was 827—656 males and 216 females, at a cost, including salaries, of £88,229 11s. 8d., being an average of £3 4s. 2d. each.

THE last advices state that the ravages of small-pox at Cape Town are worse than ever. Two thousand cases are reported, of which 600 have been fatal, mostly among natives, but the disease is spreading among the whites.

DR. ARDOVIN, French physician, attached to the General European Catholic Hospital, Alexandria, has been named by the French Government, Chevalier of the Legion of Honour, in recognition of his services during the present Egyptian crisis.

A CROWDED meeting of the St. John's Ambulance Association was held at Swanage last week, to inaugurate a centre for the Isle of Purbeck. Dr. Sieveking attended as a deputation from the Central Executive Committee, and delivered an address. It was resolved to open classes without delay, and to ask the Earl of Eldon to become President.